UNDERGRADUATE RESEARCH EXPERIENCE (UREx)

PROJECT REPORT

Wearable Fabric Electronics

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Abstract

Wearable electronics have gained significant traction as an interface between technology and human interaction, with applications spanning healthcare, communication, and human-computer interaction. However, traditional wearable devices are limited by their rigid structures and discrete designs, which restrict their scalability and functionality. This research explores a novel soldering-based methodology for constructing one-dimensional (1D) circuitry, addressing the challenges of scalability, device density, and durability in wearable fabric electronics. A 1D circuit was fabricated by connecting 96 micro-LEDs with conductive wires using a precise soldering technique. The use of a custom mold facilitated minimal spacing between components (<0.5 mm), resulting in a high device density. The circuit was successfully integrated with an Arduino microcontroller, demonstrating functionality through illumination of all 96 micro-LEDs. While the process shows potential as a scalable alternative to traditional methods, limitations such as labor intensity and structural fragility remain. Future work will focus on automating the fabrication process and enhancing material robustness. This study lays the groundwork for scalable wearable electronics, offering transformative applications in healthcare, smart textiles, and beyond.

Preface and Acknowledgements

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Introduction

Wearable devices such as smartwatches and rings have revolutionized the way we interact with technology, seamlessly integrating features like health monitoring, communication, and navigation into compact, portable form factors. Nevertheless, these devices remain limited by their small surface area, rigid structures, and reliance on specific body locations, thereby restricting their potential for large-scale physiological data collection and interaction. The future of wearable technology rather lies in fabric-based electronics that merge seamlessly with everyday clothing. Such devices promise to transform entire garments into intelligent systems capable of sensing, computing, and interacting with their surroundings.

Fabrics inherently offer advantages that make them ideal platforms for next-generation wearable electronics. They provide large surface areas for distributed sensing, flexibility for unrestricted motion, and comfort for prolonged use. By embedding electronic functionalities directly into fibers, fabrics can serve as unobtrusive, scalable systems for applications ranging from health monitoring and on-body communication to human-machine interfaces and environmental sensing.

A critical challenge in realizing fabric-based electronics lies in developing scalable and robust methods for integrating electronic components into fibers. One-dimensional (1D) circuitry, where components are arranged linearly along conductive pathways, offers a promising route to achieving this vision. This research project introduces a novel approach to fabricating scalable 1D circuits.

Aims

The aim of this research is to develop a scalable and robust methodology for fabricating one-dimensional (1D) circuitry, focusing on the integration of micro light-emitting diodes (micro-LEDs) with conductive wires in a compact and aligned configuration. By achieving precise interconnections and minimizing inter-component spacing, this study seeks to address the challenges associated with creating linear, high-density circuits suitable for integration into wearable electronic fabrics. Ultimately, this research aims to lay the groundwork for intelligent fabric-based systems, enabling applications in wearable technology, health monitoring, and advanced human-machine interfaces.

Research Question

We therefore ask, what techniques can be employed to reliably construct scalable 1D circuits by connecting micro-LEDs with conductive wires, ensuring consistent alignment and compact spacing, to enable their integration into advanced wearable electronic textiles?

The following sections detail the methodology employed, the findings of this study, and the limitations of the current approach. A comprehensive discussion of future directions is also presented, outlining the steps necessary to refine and scale this innovative technology.

Literature Review

Wearable electronics have seen rapid advancements as researchers explore new ways to integrate electronics into textiles. Numerous studies have focused on scalable methods to achieve this integration, each contributing to the evolving field of smart fabrics.

Rein et al. (2018) introduced a thermally drawn fiber process that incorporates prefabricated semiconductor diodes into fibers. Their work achieved notable breakthroughs, including embedding hundreds of discrete diodes into a single fiber while maintaining electrical connectivity. By constructing macroscopic preforms with embedded diodes and conductive wires, they created fibers capable of emitting light and performing optical communication tasks. This approach demonstrated durability, with fibers retaining functionality through machine washing and textile weaving. However, the reliance on thermal drawing introduces limitations, such as material compatibility constraints and challenges in controlling inter-device distances, which hinder its adaptability for customizable device configurations.

Similarly, Loke et al. (2019, 2021) advanced thermally drawn fiber techniques, focusing on multi-material designs for energy storage, sensing, and computational tasks. These approaches showcased the potential of integrating diverse functionalities into textiles but faced similar scalability challenges due to the fixed component spacing dictated by the preform design and the inherent labor intensity of the process.

Building on these advancements, MIT researchers demonstrated programmable fibers containing sensors, memory, and machine-learning capabilities (Ham, 2021). These fibers represent a leap toward “fiber computers” that could autonomously analyze and respond to environmental and physiological data.

This project introduces a soldering-based methodology as an alternative to thermal drawing techniques. Unlike previous methods, this approach allows for precise alignment of micro-LEDs and flexible spacing adjustments, overcoming limitations in adaptability and component density. By simplifying wire arrangements and reducing reliance on preform designs, the proposed method serves as a foundation for more scalable and robust wearable fabric electronics. However, like its predecessors, this method remains a stepping stone, requiring further development to address challenges in automation and durability, which will be critical for its adoption in real-world applications.

Methodology

This research project utilises a method that departs from thermal drawing and incorporates soldering of wires to micro-LEDs in order to create a fundamental 1D circuit.

One fiber consists of 96 micro-LEDs arranged in a straight line. A mold is used to arrange the micro-LEDs, with slots the same dimension as the micro-LEDs with negligible tolerance so that the micro-LEDs can slot in comfortably. Moreover, this method allows the micro-LEDs to be spaced extremely close to each other (>0.5mm), thus providing a high device density.

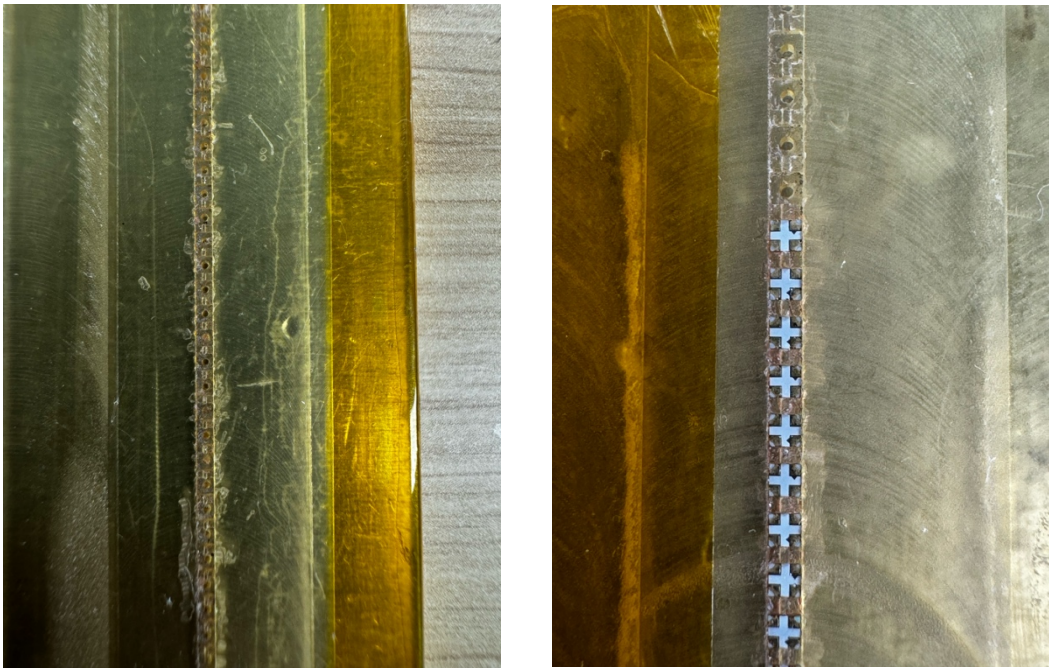


Figure 1: Close-up of mold (left) and micro-LEDs inserted in the mold

Each micro-LED is placed at an orientation 90° anticlockwise from the other, in order to allow the wires to remain straight and facilitate easier connections. A singular micro-LED consists of four connections: digital out (D_out) pin, digital in (D_in) pin, ground (G) pin, and voltage in (V_in) pin. As such, four wires will run through each micro-LED, with one wire for each connection. Rotating alternating micro-LEDs ensures that all the ground pins and V_in pins are on the same column while the D_in and D_out pins still alternate between columns. This is important because one of the four wires connects the V_in pins for all the micro-LEDs, one connects all the ground pins, and the other two connect the D_in of one micro-LED to the D_out of the other micro-LED. These wires have to be cut after two micro-LEDs have been connected to ensure a local connection and prevent short-circuiting.



Figure 2: Labelled diagram of a single micro-LED

Furthermore, it is crucial to use the correct wires and not just any wire for the connections. Certain specific criteria must be followed:

1. The wires must be flexible and be able to resist continuous deformation
2. The wires must have a coating that is easy to scrape of using a needle
3. The coating of the wire must nevertheless be able to withstand high temperatures from a soldering iron

Once the wires have been procured and the micro-LEDs have been placed into the mold, it is time to start connecting both components. The micro-LEDs also have a coating at each pin which needs to be scraped off before the wires are soldered to it. Given the microscopic size of the components, it is recommended to scrape off the coatings for each pin only after the previous one has been soldered. This is to prevent the solder from also sticking to the other pin due to its metallic surface being revealed.

Similarly, the wire coating is scraped, again keeping in mind not to scrape too much to avoid unintended connections and the possibility of short-circuiting. The wire is then first soldered to the D_in pin. Due to the miniature size of the connections and the process in general, soldering is carried out with the aid of a microscope, and a special soldering paste is used. The soldering paste used in this research project was an AG-BI-SN alloy and was applied onto the connections using a syringe. For better precision, the soldering rod also had an extension that made the tip narrower than 1mm.

Next, the same procedure is repeated for the D_out pin of the next micro-LED and so on for all subsequent micro-LEDs. It is recommended to first connect all the digital pin connections for one column together and make the cuts to isolate the D_in and D_out pairs thereafter. This is to make the process less time consuming as soldering a new piece of wire requires significantly more precision than continuing to solder the wire already in progress.

Repeat the same process for the other digital wire column, and then the ground and V_in wires. No cutting is required is required for the V_in and ground wires, and a wire length should be chosen such that there is a considerable length of wire remaining after all the micro-LED connections are made so that it is easier to connect to the microcontroller (Arduino is used for this research project).

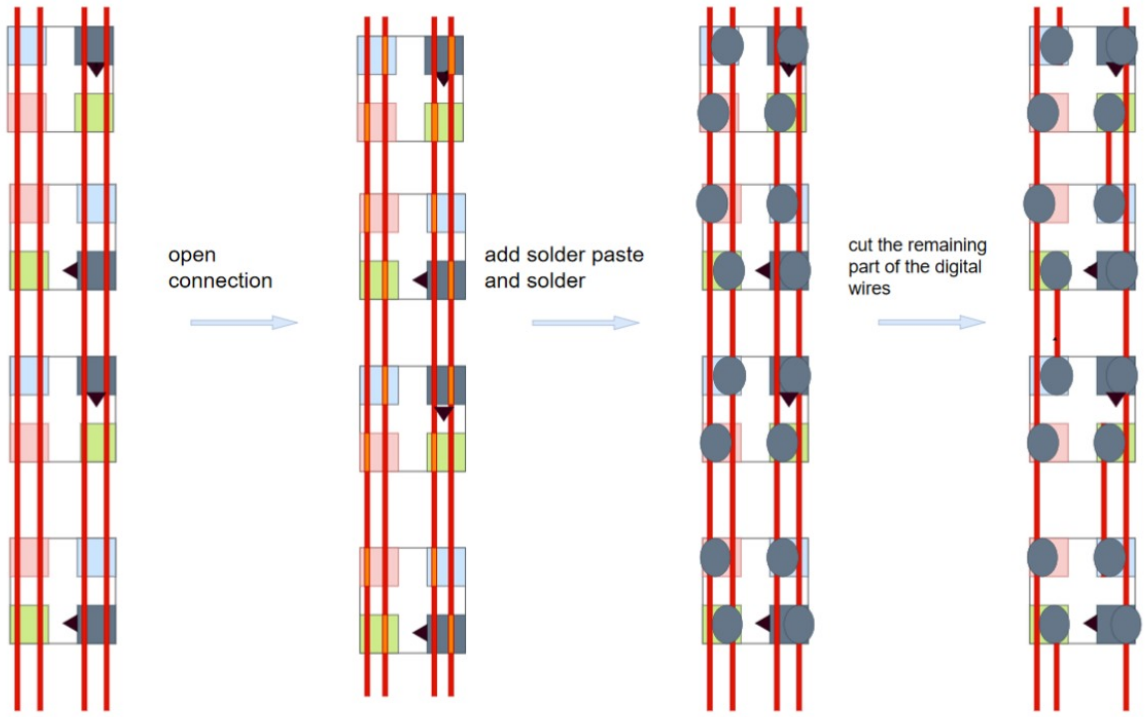


Figure 3: Schematic of the wire micro-LED connection process

Once the fiber is ready, there will be two wires exiting from the first micro-LED – the ground and V_in wires – and one wire from the last micro-LED, the D_in wire. There is no D_out wire exiting because the penultimate D_in pin is connected to the final D_out pin whereas the last D_in pin does not have a connection.

These three wires are then connected to the microcontroller's voltage, ground, and digital pin wires. This is done using a crocodile clip and soldering the ends of the exiting wires, but this time using a traditional solder paste with a higher thickness. After testing that all connections are secure, the micro-LEDs are removed from the mold by pricking each one using a thin needle from the hole at the bottom of the mold. The end result is a 1D circuit.

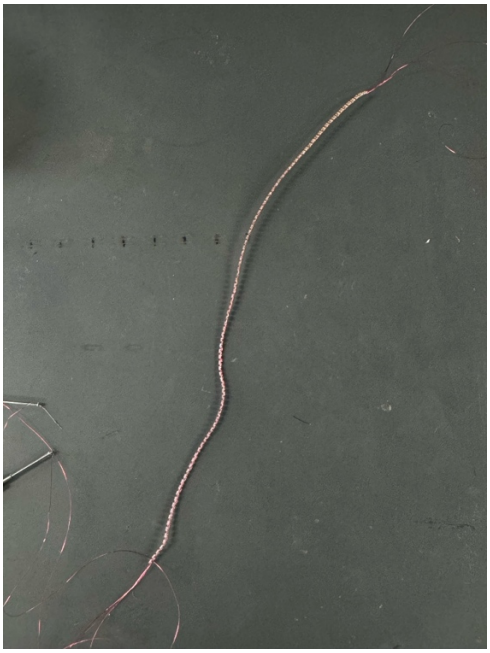


Figure 4: Finished 1D circuit consisting of 96 micro-LEDs

Findings

The research successfully demonstrated the fabrication and functionality of a 1D circuit consisting of 96 micro-LEDs soldered onto conductive wires. The circuit, upon connection to an Arduino microcontroller, was able to light up all 96 micro-LEDs, validating the integrity of the design. This achievement highlights the effectiveness of the soldering-based approach in creating high-density electronic fibers with a compact configuration.

A key finding was the significantly higher device density achieved compared to conventional thermal drawing methods. By minimizing the spacing between micro-LEDs to less than 0.5 mm, this method allows for more components to be integrated into a single fiber without compromising functionality. This increase in device density is critical for applications requiring compact and efficient systems.

The straight wire arrangement adopted in this method also simplified the fabrication process. Unlike methods requiring angular connections, which can introduce alignment complexities and increase production time, the use of aligned and parallel wires reduced the potential for errors and short circuits. This simplified configuration not only enhances reliability but also opens the door for further automation in the assembly process.

Additionally, the use of specialized soldering paste and the precise placement of components ensured robust electrical connections, which were critical for maintaining consistent functionality across all 96 micro-LEDs. The choice of an AG-BI-SN solder alloy with high thermal resilience further contributed to the stability of the circuit during fabrication.



Figure 5: Fully-functioning 1D Circuit

Limitations

The current fabrication method requires meticulous attention to detail, particularly during the soldering of wires to the micro-LEDs. Each step, including scraping off coatings, positioning micro-LEDs in the mold, and soldering under a microscope, demands a high level of manual precision. Moreover, dealing with such miniscule apparatuses further makes the process difficult to automate as mainstream products will not be feasible. This reasons make the process time-consuming, error-prone, and difficult to scale. Nevertheless, developing apparatus to precisely carry out the steps does not seem beyond the realm of possibilities and can definitely be looked into once the method is further concretized, especially given the repetitive nature of the steps.

Another limitation that was discovered during testing was the possibility of short-circuiting and its unfixability. The close spacing of micro-LEDs and the manual scraping of wire and pin coatings increase the likelihood of unintended connections. Any mistake in removing the coatings or applying solder led to short circuits, compromising the entire circuit. This was further detrimental because a short-circuited micro-LED meant that it would have to be removed from the fiber and all the connections would have to be redone. Subsequently, redoing the connections would mean adding more wire to the solder, thereby making the soldered connection even more prone to short-circuiting. Due to the microscopic nature of the components, it was also difficult to detect incorrect solders or connections. A method to reduce human error, such as improved insulation techniques or machine-assisted soldering, could mitigate this risk.

Lastly, we realised that strength and durability of the fabricated circuit were directly tied to the wire material used. Although the wires were chosen keeping in mind flexibility and, the circuit remains delicate. Repeated bending or deformation may weaken connections, especially in wearable or dynamic applications. Developing more robust materials or embedding the wires and micro-LEDs into a protective substrate could enhance durability and make the fibre more suitable for day-to-day applications.

While the developed fabrication method demonstrates significant potential in achieving high-density 1D circuitry, its limitations in scalability, susceptibility to short-circuiting, and structural fragility highlight areas requiring further refinement. Addressing these challenges through automation, improved insulation, and enhanced material robustness will be crucial for transitioning this approach from a prototype stage to practical, large-scale

applications. These improvements could pave the way for more reliable, durable, and adaptable solutions in the field of wearable electronics.

Further Directions

To refine and expand upon this research, several future directions can be explored to address existing limitations and unlock the full potential of the developed methodology.

Firstly, integration of machine-assisted fabrication presents a promising avenue to enhance scalability and minimize human error. The current manual soldering and micro-LED placement processes are labor-intensive and prone to inconsistencies, making large-scale production challenging. Implementing robotic systems equipped with precision tools for soldering and alignment could standardize the process, ensuring uniformity and reducing the time required for assembly. Such systems could also include automated diagnostic features to detect and correct errors in real-time, further improving efficiency and reliability.

Secondly, material innovations could significantly improve the performance and durability of the fabricated circuits. Advanced soldering alloys with lower melting points and enhanced bonding strength could reduce heat stress on micro-LEDs, preserving their functionality during assembly. Similarly, wires with optimized coatings that resist thermal degradation, maintain flexibility, and offer higher tensile strength could enhance the structural integrity of the circuits. Exploring such materials would not only address current durability concerns but also expand the range of environments and applications for these circuits.

Furthermore, the development of functional prototypes with additional capabilities represents an exciting step forward. Integrating sensors, data storage, or wireless communication modules into the circuits could transform them into multifunctional systems suited for diverse applications. For example, embedding physiological sensors could enable health-monitoring fabrics, while incorporating data transmission capabilities would pave the way for advanced wearable communication devices. These prototypes would provide valuable insights into the practical challenges and opportunities of deploying such technologies in real-world scenarios.

Lastly, testing in dynamic environments is essential to evaluate the circuit's performance under conditions that mimic real-world usage. Wearable electronics are often subjected to mechanical stresses such as bending, stretching, and repeated motion. Simulating

these conditions in a controlled setting would reveal the circuit's resilience and identify potential failure points. This testing would also guide the development of protective encapsulation techniques or flexible substrates to improve durability without compromising functionality.

By pursuing these directions, the methodology developed in this research can be refined into a scalable and robust framework for the fabrication of intelligent, wearable electronic systems, opening new possibilities for innovation across multiple industries.

Conclusion

This research project represents a foundational step in the journey toward developing scalable and robust methodologies for fabricating one-dimensional (1D) circuitry. By departing from traditional thermal drawing techniques and employing soldering as a method to interconnect micro-LEDs with conductive wires, we have demonstrated a practical approach to creating high-density circuits. The success of the fabricated fiber in achieving functionality—lighting 96 micro-LEDs upon connection to an Arduino microcontroller—validates the effectiveness of this method. However, as with any innovative approach, the study opens avenues for further exploration and refinement.

The primary achievement of this method lies in its ability to significantly enhance device density while simplifying the wire arrangement. The spacing between micro-LEDs, less than 0.5 mm, is a notable improvement over traditional methods, providing a clear pathway for the development of compact and efficient electronic fibers. Additionally, the method's reliance on straight wire configurations, as opposed to angular connections, simplifies the fabrication process and reduces the likelihood of alignment errors. Nevertheless, the labour-intensive nature of the process present challenges that need to be addressed before this approach can be widely adopted.

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